Gurkaran Singh

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EDUCATION

Avon High School | Avon, IN

- Expected Graduation: May 2027
- GPA: 4.4 (Weighted) / 3.9 (Unweighted) | SAT: 1380
- Advanced Coursework: AP Physics C, AP Calculus BC
- Dual-Credit: Engineering Design via Ivy Tech Community College (DESN-113: 2D CAD)

TECHNICAL SKILLS

- Software: Autodesk Inventor Professional, Autodesk Fusion 360, Flow5, XFLR5, GitHub.
- Engineering Concepts: 3D Parametric Modeling, Assembly Constraints, Mechanical Graphics, Digital Twin Validation, Aerodynamic Performance Simulation.
- Languages: Punjabi (Native), Spanish (Beginner).

INDEPENDENT ENGINEERING PROJECTS

Rotation Lab Validation Study | Spring 2026

- Developed a high-fidelity digital twin of a physical rotational apparatus using Autodesk Inventor to verify theoretical moments of inertia.
- Utilized precise calipers measurements and verified PVC density to achieve a mass calculation within a 2.7% error margin.
- Engineered a custom T-joint with smooth fillets to demonstrate Design for Manufacturability (DFM) and account for precise mass distribution.

LEADERSHIP & INVOLVEMENT

National Honor Society | Member | May 2026 – Present

- Recognized for academic excellence and character.

Distributive Education Clubs of America (DECA) | Member | August 2023 – May 2025

- Competed in business simulations through individual role-plays and written exams, demonstrating rapid problem-solving in real-world scenarios.
- Participated in community service projects to support local initiatives and promote civic engagement.

Awards: College Board National Recognition Award (2025 & 2026).